Chapter 14 — A Tour of the Standard Library

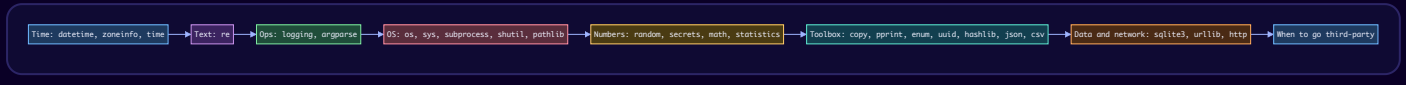
What you will learn. The modules you will reach for every week, and the traps in each: dates and time zones (`datetime` , `zoneinfo` , `time`), regular expressions in depth (`re`), professional logging (`logging`), command-line tools (`argparse`), talking to the operating system (`os` , `sys` , `subprocess` , `shutil` , `pathlib`), randomness done right (`random` vs `secrets`), numbers (`math` , `statistics` , exact `decimal` / `fractions`), a toolbox of small modules (`copy` , `pprint` , `enum` , `uuid` , `hashlib` , `json` , `csv`), an embedded database (`sqlite3`), basic HTTP (`urllib` , `http`), and when a third-party package (requests, rich, click …) is the better choice.

Prerequisites. Chapters 1–12. `collections` , `itertools` and `functools` were covered in Chapters 6, 7 and 11 — see the pointer in §14.10.

Code. Every script is in `chapters/14-standard-library-tour/code/` , uses only the standard library, needs no internet (the HTTP demo talks to a server it starts on `127.0.0.1`), and runs on Python 3.12+:

```
python chapters/14-standard-library-tour/code/01_datetime_time.py
```

"Batteries included" is Python's motto: the standard library ships with a few hundred modules. You don't need to know them all — you need to know which one to reach for and how to use it without falling into its known traps.



14.1 `datetime` and `zoneinfo` — naive vs aware

What it is

The `datetime` module has four main types:

Type	Holds	Example
`date`	year, month, day	`date(2026, 9, 28)`
`time`	hour … microsecond (no date)	`time(14, 30)`
`datetime`	date + time (+ optional time zone)	`datetime(2026, 9, 28, 14, 30)`
`timedelta`	a duration	`timedelta(days=2, hours=3)`

A datetime is **naive** if it has no time-zone info (`tzinfo is None`) — it's just "14:30 on some wall clock, somewhere". It is **aware** if it carries a `tzinfo` , so it refers to one exact instant. `zoneinfo.ZoneInfo("Asia/Kolkata")` (3.9+) gives real time-zone rules from the IANA time-zone database, including daylight saving time (**DST**). `datetime.UTC` (3.11+, alias of `timezone.utc`) is the UTC zone.

Analogy: a naive time is a note saying "meet at 9"; an aware time is "meet at 9 Tokyo time".

Why it matters

Time bugs are notoriously common: meetings an hour off after a DST change, reports that miss a day because a server runs in UTC and a user in IST, or crashes from comparing naive and aware values.

How it works

1. **Store and compute in UTC; convert to local time only for display.**
2. Create aware values: `datetime.now(UTC)` , or `datetime(..., tzinfo=ZoneInfo("Europe/London"))` . Avoid `datetime.utcnow()` — it returns a *naive* value and is deprecated since 3.12.

3. Convert with `.astimezone(ZoneInfo(...))` – same instant, different wall clock.
4. Parse and format: `datetime.fromisoformat("2026-09-28T09:15:00+05:30")` / `.isoformat()` for machine formats (ISO 8601); `strptime(text, fmt)` / `strftime(fmt)` for custom formats (`%Y-%m-%d %H:%M`, `%A` weekday name, `%I %p` 12-hour clock, `%Z` zone name).
5. Arithmetic: `datetime ± timedelta`, and `datetime − datetime → timedelta`. **Caution:** adding a `timedelta` to an aware datetime does *wall-clock* arithmetic in its zone – it does not adjust for a DST jump. For "exactly N hours later", convert to UTC, add, convert back.
6. Naive and aware values cannot be compared or subtracted (`TypeError`).
7. On Windows (no system time-zone database) install the `tzdata` package so `zoneinfo` works.

Script `01_datetime_time.py` (key parts):

```
naive = datetime(2026, 3, 8, 1, 30)
aware = datetime(2026, 3, 8, 1, 30, tzinfo=ZoneInfo("America/New_York"))
naive < aware                                  # TypeError

later = aware + timedelta(hours=1)             # wall-clock arithmetic
real = (aware.astimezone(UTC) + timedelta(hours=1)).astimezone(aware.tzinfo)

meeting_utc = datetime(2026, 9, 28, 14, 0, tzinfo=UTC)
for tz in ["Asia/Kolkata", "Europe/London", "America/Los_Angeles"]:
    print(f"  {tz:<20}", meeting_utc.astimezone(ZoneInfo(tz)).strftime("%a %H:%M %Z"))
```

Output:

```
naive tzinfo      : None
aware             : 2026-03-08T01:30:00-05:00
compare naive/aware: can't compare offset-naive and offset-aware datetimes
aware + 1h (wall) : 2026-03-08T02:30:00-05:00
aware + 1h (real) : 2026-03-08T03:30:00-04:00
  Asia/Kolkata         Mon 19:30 IST
  Europe/London        Mon 15:00 BST
  America/Los_Angeles  Mon 07:00 PDT
fromisoformat     : 2026-09-28 09:15:00+05:30 | utcoffset 5:30:00
strptime          : 2026-09-28 18:05:00 | strftime: 2026-09-28 06:05 PM
date + 30 days    : 2026-03-02 | weekday Monday
days until 2027   : 95
```

On 8 March 2026, New York clocks jump from 02:00 straight to 03:00. The "wall" result, 02:30, is a time that never existed on a New York clock; the UTC round-trip gives the correct 03:30 EDT.

Common mistakes

- Using naive datetimes in a server that talks to users in several zones.
- Hard-coding offsets (`+05:30`) instead of zone names – offsets change with DST and with politics; the zone name's rules are updated.
- Using `pytz` in new code – its `localize()` API is error-prone; `zoneinfo` replaced it.
- Expecting `date(2026, 1, 31) + "1 month"` – `timedelta` has no months (months vary in length). Use `dateutil.relativedelta` (third-party) for calendar months.

Real-world examples

Databases such as PostgreSQL store `timestamptz` as UTC; Django (`USE_TZ = True`) and most APIs exchange ISO 8601 strings in UTC; calendar apps store UTC plus the event's zone name so recurring meetings follow DST.

In the interview

"I keep all datetimes aware and in UTC internally, convert to the user's `ZoneInfo` zone only at display time, and serialise as ISO 8601. Naive datetimes are a bug source because they're ambiguous and can't be compared with aware ones."

- *Why not `utcnow()`?* It returns a naive value that looks local; use `datetime.now(UTC)`.

Self-check

1. What makes a datetime aware? → A non-None `tzinfo`.
2. How do you show a UTC time in Tokyo? → `dt.astimezone(ZoneInfo("Asia/Tokyo"))`.
3. Does `aware + timedelta(hours=1)` account for DST jumps? → No, it's wall-clock arithmetic; go via UTC.

TL;DR

Aware + UTC inside, `ZoneInfo` at the edges, ISO 8601 on the wire. `timedelta` has no months.

14.2 `time` — timestamps and timers

What it is

The `time` module is the lower-level clock interface.

Function	Returns	Use for
`time.time()`	seconds since the Unix epoch (1970-01-01 UTC), float	timestamps (can jump if the system clock is changed)
`time.perf_counter()`	highest-resolution counter, arbitrary start	measuring short durations
`time.monotonic()`	never goes backwards, arbitrary start	timeouts, rate limiters
`time.sleep(s)`	–	pause the thread
`time.time_ns()`, `perf_counter_ns()`	integer nanoseconds	avoid float precision loss

Convert a timestamp to a datetime with `datetime.fromtimestamp(ts, tz=UTC)` – e.g. `1_790_000_000` → `2026-09-21 14:13:20+00:00`.

Common mistakes

Measuring durations with `time.time()`: if NTP (the network time service) adjusts the clock mid-measurement, you get wrong or even negative durations. Use `perf_counter()` or `monotonic()`. For micro-benchmarks use the `timeit` module, which repeats the code and disables garbage collection during timing.

TL;DR

`time()` for "when", `perf_counter()` for "how long", `monotonic()` for timeouts.

14.3 `re` — regular expressions, deeper

Chapter 4 introduced `re` (`search`, `match`, `findall`, basic patterns). This section adds the parts used in real parsing work.

What it is

A **regular expression** (regex) is a small pattern language for matching text. Python's `re` module compiles a pattern into a matcher object and runs it with a backtracking engine.

How it works — the essentials

Feature	Syntax	Meaning
Raw strings	`r"\d+"`	always use them so `\` reaches `re` unchanged
Group	`( … )`	capture a part; `m.group(1)` or `m[1]`
Named group	`(?P<year>\d{4})`	`m["year"]`, `m.groupdict()`
Non-capturing	`(?: … )`	group without capturing
Alternation	`cat|dog`	either
Quantifiers	`* + ? {m,n}`	greedy by default
Lazy	`*? +? ??`	match as little as possible
Anchors	`^ $ \b \A \Z`	line start/end, word boundary, string start/end
Lookahead	`(?=…)`, `(?!…)`	require / forbid what follows, without consuming
Lookbehind	`(?<=…)`, `(?<!…)`	same, for what precedes (fixed width)
Backreference	`\1`, `(?P=name)`	the same text again

Functions: `re.search` (anywhere), `re.match` (only at the start), `re.fullmatch` (entire string — use it for validation), `re.findall` (list of strings, or tuples if there are groups), `re.finditer` (iterator of match objects — memory-friendly and gives positions), `re.sub(pattern, repl, s)` (repl may be a string with `\1` or a **function** of the match), `re.split`.

Flags: `re.IGNORECASE` (`re.I`), `re.MULTILINE` (`^` / `$` match at each line), `re.DOTALL` (`.` also matches newline), `re.VERBOSE` (`re.X`: whitespace and `#` comments allowed inside the pattern).

Compile once: `LINE = re.compile(...)` at module level. `re` also caches recently used patterns, so this is mainly for readability and reuse.

Script `02_regex_deeper.py` parses log lines with a verbose, named-group pattern:

```
LINE = re.compile(r"""
    ^(?P<date>\d{4}-\d{2}-\d{2})\s     # date
    (?P<time>[\d:]{8})\s+
    (?P<level>[A-Z]+)\s+
    \[(?P<svc>\w+)\]\s                 # service in brackets
    (?P<msg>.*?)\s                     # lazy: stop before the last field
    user=(?P<user>\d+)$
""", re.VERBOSE | re.MULTILINE)

for m in LINE.finditer(LOG):
    if m["level"] == "ERROR":
        print("error:", m.group("svc", "user"), "|", m["msg"])

re.sub(r"\d{12}(\d{4})", lambda m: "*" * 12 + m[1], card)   # function replacement
re.sub(r"(\w+), (\w+)", r"\2 \1", "Lovelace, Ada")          # backreferences
strong = re.compile(r"(?=.*\d)(?=.*[A-Z]).{8,}")            # two lookaheads
```

Output:

```
error: ('payments', '42') | card declined
error: ('auth', '7') | bad password
findall groups  : ['42', '7', '7']
greedy          : <a><b>
lazy            : <a>
masked          : pay ***********1111 now
swap            : Ada Lovelace
strong('password'  ): False
strong('Passw0rdX' ): True
IGNORECASE      : ['ERROR', 'ERROR']
split multi     : ['a', 'b', 'c ', 'd']
match vs search : None 1
```

Trade-offs / when not to use it

- **Structured formats have parsers**: JSON → `json`, CSV → `csv`, HTML → an HTML parser (BeautifulSoup/lxml), URLs → `urllib.parse`, emails → a validation library. Regex on these breaks on edge cases.
- Simple checks are clearer with string methods: `s.startswith("http")`, `"@" in s`, `s.isdigit()`.
- **Catastrophic backtracking**: nested quantifiers like `(a+)+$` on a long non-matching input take exponential time. This caused real outages (Stack Overflow in 2016, Cloudflare in 2019). Avoid nested/ambiguous quantifiers on untrusted input; since 3.11 `re` supports atomic groups `(?>…)` and possessive quantifiers `*+` to prevent backtracking.

Common mistakes

- Forgetting the `r` prefix (`"\b"` is a backspace character, not a word boundary).
- Using `match` when you meant `search` (or `fullmatch` for validation).
- Greedy `.*` swallowing too much — use `.*?` or a negated class like `[^\]]*`.
- `findall` with groups returns only the groups, not the whole match.

In the interview

"I compile patterns with named groups and `re.VERBOSE` so they're readable, use `finditer` for large inputs and `fullmatch` for validation, and I avoid regex for structured formats. I'm careful about nested quantifiers because of catastrophic backtracking on untrusted input."

TL;DR

Raw strings, named groups, `finditer`, `fullmatch`, `sub` with a function, `VERBOSE` for readability, lazy quantifiers when greedy grabs too much, and no regex for JSON/HTML/CSV.

14.4 `logging` — levels, handlers, formatters, best practice

What it is

`logging` is the standard way for programs and libraries to report what they are doing. Four building blocks:

Part	Job	Example
Logger	the object your code calls; named hierarchically with dots	`logging.getLogger("shop.orders")`
Handler	sends records somewhere	`StreamHandler` (console), `FileHandler`, `RotatingFileHandler`, `TimedRotatingFileHandler`, `SysLogHandler`, `QueueHandler`
Formatter	turns a record into text	`"%(asctime)s %(levelname)s %(name)s %(message)s"`
Filter	drops or edits records	add a request ID to each record

Levels (numbers in brackets): `DEBUG` (10) detail for developers · `INFO` (20) normal events · `WARNING` (30) something unexpected, still working · `ERROR` (40) an operation failed · `CRITICAL` (50) the program may not continue.

Why it matters (logging vs print)

`print` has no levels, no timestamps, no destinations, and cannot be switched off per module. With logging, the **same code** can be silent in tests, show INFO on the console in development, and send JSON to a log collector in production — only the configuration changes.

How it works

1. Each module does `log = logging.getLogger(__name__)`. Names form a tree: `shop.orders` → `shop` → root.
2. A call like `log.info("order placed qty=%d", qty)` creates a **LogRecord** if the logger's effective level allows INFO.
3. The record goes to the logger's handlers, then **propagates** up to the parent's handlers, and so on to the root.
4. Each handler has its own level and formatter, so one record can go to the console (INFO+) and a file (DEBUG+) in different formats.
5. With no configuration at all, WARNING and above are printed to stderr by a fallback handler ("last resort"); DEBUG/INFO are dropped because the root logger's default level is WARNING.
6. Configure **once, at program start** (in `main`), with `logging.basicConfig(level=..., format=...)` for simple apps or `logging.config.dictConfig({...})` for real ones. `basicConfig` does nothing if the root logger already has handlers, unless you pass `force=True`.

Script `03_logging_demo.py`:

```
log = logging.getLogger("shop.orders")      # normally getLogger(__name__)

def place_order(qty: int) -> None:
    log.debug("validating qty=%s", qty)
    if qty <= 0:
        log.warning("rejected order with qty=%d", qty)
        return
    try:
        1 / (qty - 3)
    except ZeroDivisionError:
        log.exception("pricing failed for qty=%d", qty)   # ERROR + traceback
        return
    log.info("order placed qty=%d", qty)

logging.config.dictConfig({
    "version": 1,
    "disable_existing_loggers": False,
    "formatters": {"short": {"format": "%(levelname)-7s %(name)s: %(message)s"},
                   "full":  {"format": "%(asctime)s %(levelname)s %(name)s "
                                       "%(filename)s:%(lineno)d %(message)s"}},
    "handlers": {"console": {"class": "logging.StreamHandler", "level": "INFO",
                             "formatter": "short", "stream": "ext://sys.stdout"},
                 "file": {"class": "logging.StreamHandler", "level": "DEBUG",
                          "formatter": "full", "stream": file_like}},
    "loggers": {"shop": {"level": "DEBUG", "handlers": ["console", "file"]}},
})
```

Output (console handler; path shortened):

```
INFO    shop.orders: order placed qty=2
WARNING shop.orders: rejected order with qty=0
ERROR   shop.orders: pricing failed for qty=3
Traceback (most recent call last):
  File ".../03_logging_demo.py", line 15, in place_order
    1 / (qty - 3)
    ~~^~~~~~~~~~
ZeroDivisionError: division by zero
--- file handler got 11 lines; DEBUG line present: True
logger chain: shop.orders -> shop -> root
effective level: DEBUG
```

The console showed INFO and above; the "file" handler (a `StringIO` here) received the DEBUG lines too.

Best practice

1. `getLogger(__name__)` in every module; never log on the root logger directly in libraries.
2. **Libraries never configure logging.** They only create loggers (optionally adding `logging.NullHandler()`); the application decides where logs go.
3. Pass values as arguments – `log.info("user %s", uid)` – not f-strings. Formatting is then skipped when the level is disabled, and log aggregators can group identical messages.
4. Inside `except`, use `log.exception(...)` (ERROR level + traceback) or `exc_info=True`.
5. Don't log secrets or personal data (passwords, tokens, card numbers).
6. In production prefer **structured** (JSON) logs with context fields (request ID, user ID), rotated files or stdout collected by the platform (the "12-factor" approach: log to stdout, let Docker/Kubernetes collect). `structlog` and `python-json-logger` are common helpers.
7. For high-throughput or async apps, `QueueHandler` + `QueueListener` moves slow I/O off the hot path.

Common mistakes

- Calling `basicConfig` in an imported module (it silently wins or silently does nothing).
- Adding handlers inside a function called repeatedly → each message printed many times. Configure once.
- Messages appear twice: a handler on both a child logger and the root, and `propagate` left True.
- `disable_existing_loggers` defaults to `True` in `dictConfig` – loggers created before configuration are silenced. Set it to `False`.

Real-world examples

Django and Flask configure logging through `dictConfig` (Django's `LOGGING` setting); `uvicorn`, `requests` / `urllib3` and `boto3` all emit through named loggers you can tune, e.g. `logging.getLogger("urllib3").setLevel(logging.WARNING)`.

In the interview

"Each module gets `getLogger(__name__)`. The application configures handlers and formatters once with `dictConfig`. Libraries don't configure logging. I use lazy `%s` arguments, `log.exception` in except blocks, and structured JSON logs to stdout in containers."

- *Why not print?* No levels, no routing, can't be turned off per module, no timestamps or context.
- *Why are my debug logs missing?* The logger or handler level is above DEBUG – the default root level is WARNING.

Self-check

1. Default level of the root logger? → WARNING.
2. Which call logs a traceback? → `log.exception(...)` inside `except` (or `exc_info=True`).
3. Why are messages duplicated? → Handlers on several loggers in the chain with propagation.
4. Should a library call `basicConfig`? → No.

TL;DR

Logger (per module) → handlers (where) → formatters (how). Configure once in the app, lazy args, `log.exception`, structured logs in production.

14.5 `argparse` — command-line interfaces

What it is

`argparse` turns `sys.argv` (the list of words typed after the program name) into a validated object, generates `--help`, and prints usage errors.

How it works

1. Create `ArgumentParser(prog=..., description=...)`.
2. `add_argument("title")` — a **positional** argument (required, by position).
3. `add_argument("--priority", type=int, choices=..., default=...)` — an **option**; `type` converts the text, `choices` restricts values.
4. `action="store_true"` for on/off flags, `action="count"` for `-vv`, `action="append"` for repeatable options, `nargs="+"` for one-or-more values, `required=True` to force an option.
5. `add_subparsers()` for git-style subcommands (`todo add`, `todo list`).
6. `parse_args()` returns a `Namespace`; on invalid input it prints usage to stderr and exits with status `2`. `-h/--help` is added automatically.
7. Make `main(argv=None)` accept a list so tests can call it without a real command line.

Script `04_argparse_cli.py`:

```
p = argparse.ArgumentParser(prog="todo", description="A tiny todo CLI.")
p.add_argument("-v", "--verbose", action="count", default=0)
sub = p.add_subparsers(dest="command", required=True)

add = sub.add_parser("add", help="add a task")
add.add_argument("title")
add.add_argument("--priority", type=int, choices=range(1, 4), default=2)
add.add_argument("--tag", action="append", default=[])

ls = sub.add_parser("list", help="list tasks")
ls.add_argument("--done", action="store_true")
ls.add_argument("--limit", type=int, default=10)
```

Output:

```
$ todo add Buy milk --priority 1 --tag home --tag shop
  parsed: {'verbose': 0, 'command': 'add', 'title': 'Buy milk', 'priority': 1, 'tag': ['home', 'shop']}
$ todo -vv list --done
  parsed: {'verbose': 2, 'command': 'list', 'done': True, 'limit': 10}
$ todo add x --priority 9
usage: todo add [-h] [--priority {1,2,3}] [--tag TAG] title
todo add: error: argument --priority: invalid choice: '9' (choose from '1', '2', '3')
  exit code: 2
usage: todo [-h] [-v] {add,list} ...
```

(`"Buy milk"` was one list item here; in a real shell you'd quote it: `todo add "Buy milk"`.)

Making it runnable

Add a `[project.scripts]` entry in `pyproject.toml` (`todo = "todo.cli:main"`) so `pip install` creates a `todo` command, or run as a module with `python -m todo` using a `__main__.py` (Chapter 8). Finish with `raise SystemExit(main())` so the return code reaches the shell.

Trade-offs

Library	Style	Good for
`argparse` (stdlib)	imperative, no dependency	scripts, tools that must not need installs
`click`	decorators, composable command groups	larger CLIs (Flask's `flask` command uses it)
`typer`	builds a click CLI from type hints	modern typed CLIs
`sys.argv` directly	manual	only for one or two trivial arguments

Common mistakes

- `type=bool` – `bool("False")` is `True`. Use `action="store_true"`.
- Parsing `sys.argv` by hand for anything non-trivial.
- Calling `parse_args()` at import time, which breaks tests; do it in `main`.

TL;DR

Positionals + options with `type` / `choices` / `action`; subparsers for commands; `main(argv=None)` for testability; exit code 2 on usage errors.

14.6 `os`, `sys`, `subprocess`, `shutil` (+ `pathlib` recap)

What it is

Module	Area	Most-used parts
`os`	the operating system	`os.environ`, `os.getenv`, `os.cpu_count()`, `os.getpid()`, `os.walk`, `os.sep`, `os.replace` (atomic rename)
`sys`	the running interpreter	`sys.argv`, `sys.exit()`, `sys.path`, `sys.version_info`, `sys.platform`, `sys.executable`, `sys.stdin/stdout/stderr`, `sys.modules`
`subprocess`	running other programs	`subprocess.run([...], capture_output=True, text=True, check=True, timeout=...)`
`shutil`	high-level file operations	`copy2`, `copytree`, `move`, `rmtree`, `make_archive`, `which`, `disk_usage`
`pathlib`	paths as objects (Chapter 9)	`Path`, `/`, `.read_text()`, `.glob()`, `.rglob()`, `.mkdir(parents=True)`
`tempfile`	throw-away files/dirs	`TemporaryDirectory()`, `NamedTemporaryFile()`

How it works – the key rules

1. **Configuration from the environment**: `os.environ.get("APP_ENV", "dev")` (never `os.environ["X"]` if it may be missing – that raises `KeyError`). Environment values are always strings.
2. **Paths**: use `pathlib.Path` for new code; `os.path` functions still work but are more verbose.
3. **Running commands** with `subprocess.run`:
   - Pass a **list** of arguments. Avoid `shell=True` with any user-supplied text – it enables shell injection (`"file.txt; rm -rf ~"`).
   - `capture_output=True, text=True` gives `result.stdout` as a `str`.
   - `check=True` raises `CalledProcessError` on a non-zero exit code; `timeout=` raises `TimeoutExpired`.
   - Use `sys.executable` to run "the same Python that runs me".
   - `os.system` and `os.popen` are older, less safe interfaces – prefer `subprocess`.

4. **Deleting trees** with `shutil.rmtree` is permanent — double-check the path. `os.replace(tmp, final)` gives an atomic "write then swap" for safe file updates.
5. **Exit codes:** `sys.exit(0)` success, non-zero failure (it raises `SystemExit`, so `finally` blocks still run).

Script `05_os_sys_subprocess_shutil.py` (key parts):

```
with tempfile.TemporaryDirectory() as tmp:
    root = Path(tmp)
    (root / "src" / "pkg").mkdir(parents=True)
    (root / "src" / "pkg" / "a.py").write_text("print('hi')\n")
    shutil.copytree(root / "src", root / "backup")
    archive = shutil.make_archive(str(root / "bundle"), "zip", root / "src")
    shutil.rmtree(root / "backup")
    r = subprocess.run([sys.executable, str(root / "src" / "pkg" / "a.py")],
                       capture_output=True, text=True, check=True, timeout=10)
```

Output:

```
python      : (3, 14) | platform: darwin
argv[0]     : 05_os_sys_subprocess_shutil.py
HOME set    : True | APP_ENV = dev (default)
cpu_count>0 : True
copied      : ['backup/notes.txt', 'backup/pkg', 'backup/pkg/a.py']
py files    : ['a.py', 'a.py']
archive     : bundle.zip True
after rmtree: ['bundle.zip', 'src']
child stdout: hi | returncode 0
failed child: returncode 3
which python: True
disk free GB: True
```

Common mistakes

- `subprocess.run("ls " + user_input, shell=True)` — injection risk.
- Ignoring the return code (forgetting `check=True`).
- `os.chdir` in library code — it changes the working directory for the whole process.
- String-concatenating paths with `"/"` instead of `Path` or `os.path.join`.

In the interview

"I use `subprocess.run` with a list of arguments, `check=True`, a timeout, and never `shell=True` on user input. Config comes from `os.environ.get` with defaults, paths are `pathlib.Path`, and temporary work goes in `tempfile.TemporaryDirectory`."

TL;DR

`os` = OS & env, `sys` = interpreter, `subprocess.run([...], check=True)` = other programs, `shutil` = copy/move/delete trees, `pathlib` = paths.

14.7 `random` vs `secrets`

What it is

- `random` produces **pseudo-random** numbers from the Mersenne Twister algorithm: fast, reproducible with a seed, statistically good — but **predictable** once enough outputs are observed.
- `secrets` (3.6+) draws from the operating system's cryptographically secure source (`os.urandom`): unpredictable, not reproducible.

Rule: `random` for simulations, games, sampling and tests; `secrets` for passwords, tokens, keys, password-reset links.

How it works

`random`: `random.Random(42)` (your own seeded generator – better than the global `random.seed` because other code can't disturb it), `randint(a, b)` (both ends included), `random()` in [0, 1), `uniform`, `choice`, `choices(pop, weights=, k=)` (with replacement), `sample(pop, k)` (without replacement), `shuffle` (in place), `gauss(mu, sigma)`.

`secrets`: `token_bytes(n)`, `token_hex(n)` (2n hex chars), `token_urlsafe(n)` (about 1.3 × n characters, URL-safe), `choice(seq)`, `randbelow(n)`, and `compare_digest(a, b)` – a constant-time comparison that prevents timing attacks when checking tokens.

Output of `06_random_secrets_math_stats.py` (first part):

```
randint      : [6, 1, 1, 6, 3]
choice       : red
sample       : [28, 17, 94]
shuffle      : [7, 2, 5, 6, 0, 3, 4, 9, 8, 1]
weighted     : ['A', 'A', 'A', 'B', 'B', 'A', 'A', 'A']
gauss        : 1.1491
token_urlsafe: 22 chars
token_hex(8) : 16 hex chars
compare_digest: True
```

Common mistakes

- `random` for security tokens (predictable).
- Comparing secrets with `==` (can leak timing information) – use `secrets.compare_digest`.
- For NumPy work, use `numpy.random.default_rng(seed)` instead (ML handbook).

TL;DR

`random` = reproducible simulation; `secrets` = anything an attacker must not guess.

14.8 `math`, `statistics`, `decimal`, `fractions`

What it is

`math` has fast functions on single floats/ints (C library wrappers). `statistics` has descriptive statistics on small-to-medium data lists – correct and readable, but slower than NumPy/pandas for big arrays.

How it works — highlights

Need	Function	Note
Compare floats	`math.isclose(a, b)`	default relative tolerance 1e-9; `0.1 + 0.2 == 0.3` is `False`
Rounding	`math.floor`, `math.ceil`, `math.trunc`, `round`	`round` uses **banker's rounding**: `round(2.5) == 2`, `round(3.5) == 4`
Roots, distances	`sqrt`, `isqrt` (exact int), `hypot`, `dist`	
Counting	`comb(n, k)`, `perm(n, k)`, `factorial`	5 choose 2 = 10
Number theory	`gcd`, `lcm`	`lcm` since 3.9
Logs, constants	`log(x, base)`, `log2`, `log10`, `exp`, `pi`, `e`, `inf`, `nan`	
Accurate sums	`math.fsum`	exact float sum
Averages	`statistics.mean`, `fmean`, `median`, `mode`, `multimode`	
Spread	`pstdev` / `pvariance` (population), `stdev` / `variance` (sample, divides by n−1)	
Quartiles	`statistics.quantiles(data, n=4)`	
Relations	`correlation`, `covariance`, `linear_regression`	3.10+

Output (second part):

```
isclose      : False True
floor/ceil   : -3 -2 2 4
sqrt, hypot  : 4.0 5.0
comb, perm   : 10 20 120
gcd, lcm     : 6 12
log          : 3.0 3.0 2.71828
fsum vs sum  : 1.0 1.0
mean/median  : 5 4.5
mode         : 4
pstdev/stdev : 2.0 2.1381
quantiles    : [4.0, 4.5, 6.5]
correlation  : 0.7746
linear_reg   : LinearRegression(slope=0.6, intercept=2.2)
```

Worked check for the spread: data `[2, 4, 4, 4, 5, 5, 7, 9]` has mean 5; squared deviations sum to 9+1+1+1+0+0+4+16 = 32. Population variance = 32/8 = 4 → `pstdev` = 2.0. Sample variance = 32/7 ≈ 4.571 → `stdev` ≈ 2.1381.

Note on `fsum vs sum`: since Python 3.12, the built-in `sum()` of floats uses compensated (Neumaier) summation, so `sum([0.1] * 10)` is `1.0`; on 3.11 and earlier it was `0.9999999999999999`. `math.fsum` remains the exactly-rounded option.

`decimal` and `fractions` — exact arithmetic

Floats are binary, so `0.1` cannot be stored exactly and money sums drift. Two stdlib types avoid this:

```
from decimal import Decimal, ROUND_HALF_UP
from fractions import Fraction

Decimal("0.1") + Decimal("0.2")                                  # Decimal('0.3') - exact
Decimal(0.1)                                                     # 0.1000000000000000055511... (built from the
float!)
Decimal("2.675").quantize(Decimal("0.01"), rounding=ROUND_HALF_UP)   # Decimal('2.68')
round(2.675, 2)                                                  # 2.67 - the float is really 2.67499999...
Fraction(1, 3) + Fraction(1, 6)                                  # Fraction(1, 2)
```

- `Decimal` is base-10 with a configurable precision (28 significant digits by default) and explicit rounding modes. Use it for money, and always build it from **strings**, not floats.
- `Fraction` stores an exact numerator/denominator — handy for maths puzzles and exact ratios, but slow if numbers grow large.
- Both are much slower than `float`; use them where exactness matters, not in numeric loops.

Common mistakes

- `math.floor(-2.5)` is `-3` (towards minus infinity), not `-2`; `int(-2.5)` is `-2` (towards zero).
- Using `==` on floats. Use `isclose`, or `decimal.Decimal` for money.
- Mixing up `stdev` (sample) and `pstdev` (population).

TL;DR

`math` for one-number maths, `statistics` for small data summaries; `isclose` for float comparison; `round` is banker's rounding.

14.9 A toolbox of small modules: `copy`, `pprint`, `enum`, `uuid`, `hashlib`, `json`, `csv`

`copy` — shallow vs deep

- `copy.copy(x)` — a **shallow** copy: a new outer container that shares the same inner objects.
- `copy.deepcopy(x)` — copies recursively (handles cycles with a memo dict). Slower; customise with `__copy__` / `__deepcopy__`.
- `copy.replace(obj, **changes)` (3.13+) — a modified copy of a dataclass, namedtuple, datetime, etc.

```
shallow sees change: ['a', 'b', 'c'] | deep: ['a', 'b']
```

The shallow copy shares the `hosts` list, so appending through `cfg` shows up in it.

`pprint` — readable nested data

`pprint.pp(obj, width=50)` (3.8+) prints nested structures across lines, keeping insertion order (`pprint.pprint` sorts dict keys by default). Output:

```
{'service': 'api',
 'replicas': 3,
 'env': {'DB_URL': 'postgres://db:5432/app',
         'DEBUG': False},
 'ports': [80, 443],
 'owners': ['platform-team', 'sre']}
```

Also `pprint.pformat` returns the string, handy in log messages.

`enum` – named constants

Covered in Chapter 10. `Enum` for a closed set of named values, `StrEnum` (3.11+) whose members *are* strings (great for JSON and databases), `IntEnum`, `Flag` for combinable bits, `auto()` for automatic values (for `StrEnum` it gives the lower-cased member name):

```
StrEnum: active BANNED | Enum: Color.GREEN ['RED', 'GREEN']
```

`uuid` – unique identifiers

A **UUID** is a 128-bit identifier, written as 36 characters, that can be generated anywhere without coordination.

Function	Based on	Use
`uuid4()`	random	the default for IDs
`uuid5(namespace, name)` / `uuid3`	SHA-1 / MD5 hash of a name	the same input always gives the same ID
`uuid1()`	time + MAC address	leaks the machine's MAC; avoid
`uuid7()` (3.14+)	Unix time + random	time-ordered, so database indexes stay compact

```
uuid4 version: 4 | uuid5 is deterministic: 6639460f-3425-5329-8097-a58f06127860
```

`hashlib` – cryptographic hashes

A **hash function** maps any bytes to a fixed-size digest; the same input always gives the same digest, and a good one makes it infeasible to find two inputs with the same digest.

```
hashlib.sha256(b"hello").hexdigest()      # '2cf24dba5fb0a30e...'
h = hashlib.sha256()
for chunk in (b"hel", b"lo"):              # hash large files chunk by chunk
    h.update(chunk)
key = hashlib.pbkdf2_hmac("sha256", b"s3cret", salt, 100_000)
```

Output:

```
sha256('hello'): 2cf24dba5fb0a30e ...
streamed equal : True
pbkdf2 key len : 32
```

Rules:

- Use SHA-256 (or BLAKE2, `hashlib.blake2b`) for integrity checks, cache keys, content addressing (Git identifies objects by SHA-1 and is adding SHA-256 support).
- MD5 and SHA-1 are broken for security; fine only for non-security checksums.
- **Never store passwords with a plain fast hash.** Use a slow, salted key-derivation function: `hashlib.scrypt` or `pbkdf2_hmac` with a random per-user salt (`secrets.token_bytes(16)`) and a high iteration count – or the third-party `argon2-cffi` / `bcrypt`. (The demo's fixed salt is only for reproducibility.)
- For message authentication use `hmac` (`hmac.new(key, msg, "sha256")`) and compare with `hmac.compare_digest`.
- Python's built-in `hash()` is **not** a stable hash: for strings it is randomised per process.

`json` and `csv` – recap (Chapter 9)

- `json.dumps(obj, default=str, sort_keys=True, indent=2)` – `default` is called for unsupported types (dates, sets, Decimal); tuples become lists; `json.loads` returns plain dicts/lists.
- `csv.DictWriter` / `DictReader` handle quoting (`"Li, Wei"` stays one field). Open files with `newline=""`.

```
json: {"id": 7, "tags": ["a", "b"], "when": "2026-09-28"} | back: ['a', 'b']
csv text: ['name,city', 'Asha,Pune', '"Li, Wei",Oslo']
csv read: {'name': 'Li, Wei', 'city': 'Oslo'}
```

More small modules worth knowing

Module	Does	Example
`tomllib` (3.11)	Read TOML config files (read-only)	`tomllib.load(open("pyproject.toml", "rb"))`
`zipfile`, `tarfile`, `gzip`	Read/write archives and compressed files	`gzip.open("log.gz", "rt")` streams a compressed text file
`base64`	Encode bytes as ASCII text	`base64.b64encode(b"hi")` → `b'aGk='`
`textwrap`	Wrap/indent/dedent paragraphs	`textwrap.dedent(help_text)`
`shlex`	Split a command line like a shell does	`shlex.split('git commit -m "fix bug"')` → `['git', 'commit', '-m', 'fix bug']`
`difflib`	Diffs and "did you mean" matches	`difflib.get_close_matches("pyhton", ["python", "java"])`
`getpass`	Read a password without echoing it	`getpass.getpass()`
`string`	Constants and simple templates	`string.ascii_lowercase`, `string.Template("$name")`
`timeit`, `cProfile`, `tracemalloc`	Timing, profiling, memory tracing	Chapter 17

TL;DR

`deepcopy` for independent nested copies; `pprint.pp` for readable dumps; `StrEnum` for string constants; `uuid4` / `uuid7` for IDs; SHA-256 for integrity, scrypt/pbkdf2/argon2 for passwords; `json` `default=`; csv via `DictReader/DictWriter`.

14.10 Pointer: `collections`, `itertools`, `functools`

These three are covered in depth elsewhere; here is where to look:

Module	Highlights	Chapter
`collections`	`Counter`, `defaultdict`, `deque`, `namedtuple`, `OrderedDict`, `ChainMap`	6
`itertools`	`chain`, `islice`, `groupby`, `product`, `permutations`, `combinations`, `accumulate`, `batched` (3.12), `pairwise`	11
`functools`	`lru_cache` / `cache`, `partial`, `reduce`, `wraps`, `total_ordering`, `singledispatch`, `cached_property`	7, 11, 12
`heapq`, `bisect`	priority queues, binary search in sorted lists	5, 6
`dataclasses`, `abc`, `contextlib`	records, abstract classes, context managers	10, 12

14.11 `sqlite3` — an embedded SQL database

What it is

SQLite is a complete SQL database stored in a single file (or in memory), running inside your process — no server to install. `sqlite3` is Python's built-in driver for it (following the DB-API 2.0 standard, PEP 249, shared with PostgreSQL/MySQL drivers).

How it works

1. `con = sqlite3.connect("app.db")` (or `":memory:"`).
2. `con.execute(sql, params)` runs one statement; `executemany` runs it for each row.
3. **Always use placeholders** (`?` or `:name`) for values — the driver escapes them, preventing **SQL injection**. Never build SQL with f-strings from user input.
4. `with con:` wraps a **transaction**: it commits on success and rolls back if an exception escapes (it does *not* close the connection — call `con.close()`).
5. `con.row_factory = sqlite3.Row` lets you read columns by name.
6. Results: `.fetchone()`, `.fetchall()`, or iterate the cursor.

Script `08_sqlite_demo.py`:

```
con = sqlite3.connect(":memory:")
con.row_factory = sqlite3.Row
con.execute("""CREATE TABLE expense (
                id INTEGER PRIMARY KEY,
                category TEXT NOT NULL,
                amount REAL NOT NULL CHECK (amount > 0))""")
with con:
    con.executemany("INSERT INTO expense (category, amount) VALUES (?, ?)", rows)

total = con.execute("SELECT SUM(amount) AS s FROM expense WHERE category = ?",
                    (cat,)).fetchone()["s"]

try:
    with con:                           # both inserts, or neither
        con.execute("INSERT INTO expense (category, amount) VALUES ('gift', 5)")
        con.execute("INSERT INTO expense (category, amount) VALUES ('bad', -1)")
except sqlite3.IntegrityError as e:
    print("rolled back:", e)
```

Output:

```
food total: 19.75
  travel  n=1 sum=40.0
  food    n=2 sum=19.75
  books   n=1 sum=18.0
rolled back: CHECK constraint failed: amount > 0
rows now: 4
```

The valid `gift` insert was rolled back together with the invalid one — that is atomicity.

When not to use it

Many concurrent writers (SQLite allows one writer at a time; WAL mode helps readers) or a database shared by several servers over a network — use PostgreSQL/MySQL. SQLite is excellent for apps, tests, caches, CLIs, prototypes and small-to-medium websites.

Real-world examples

SQLite is in every Android and iOS device, in browsers (Firefox and Chrome store history/bookmarks in it) and in many desktop apps. Django uses it as the default development database.

Common mistakes

- `f"... WHERE name = '{name}'"` – injection.
- Forgetting to commit (changes vanish when the connection closes).
- `(cat)` instead of `(cat,)` – parameters must be a sequence; a one-item tuple needs the comma.

TL;DR

`connect` → `execute` with `?` placeholders → `with con:` for transactions → `Row` for named columns.

14.12 `urllib` and `http` (brief)

What it is

- `urllib.parse` – split, build and encode URLs: `urlparse`, `parse_qs`, `urlencode`, `urljoin`, `quote`.
- `urllib.request` – a basic HTTP client: `urlopen(Request(url, headers=..., data=...), timeout=...)`.
- `urllib.error` – `HTTPError` (server replied with 4xx/5xx) and `URLError` (couldn't connect).
- `http.HTTPStatus` – named status codes (`HTTPStatus.NOT_FOUND == 404`).
- `http.server` – a minimal HTTP server (also `python -m http.server 8000` to share a folder). The docs warn it is **not for production**.

Script `09_urllib_http.py` starts a local server on `127.0.0.1` (port 0 = "any free port") in a background thread, then calls it:

```
u = urlparse("https://api.example.com:8443/v1/users?id=7&sort=name#top")
print("parsed :", u.scheme, u.hostname, u.port, u.path, parse_qs(u.query))

req = Request(f"{base}/greet?{urlencode({'name': 'Asha'})}",
              headers={"Accept": "application/json"})
with urlopen(req, timeout=5) as resp:
    print("GET    :", resp.status, resp.headers["Content-Type"], json.load(resp))
```

Output:

```
parsed : https api.example.com 8443 /v1/users {'id': ['7'], 'sort': ['name']}
encode : q=type+hints&page=2
urljoin: https://x.org/docs/b
GET    : 200 application/json {'message': 'hello Asha'}
error  : 404 Not Found
```

When not to use it

For real HTTP clients, `urllib.request` lacks connection pooling, convenient JSON/auth/retry handling and async. Use **requests** (synchronous, the most popular) or **httpx** (sync + async, HTTP/2). For servers, use a framework (FastAPI, Flask, Django) behind a production server (uvicorn, gunicorn).

Common mistakes

- No `timeout=` – a stuck server hangs your program forever.
- Building query strings by hand instead of `urlencode` (spaces and `&` break).

TL;DR

`urllib.parse` is great; `urllib.request` is fine for quick, dependency-free scripts; use requests/httpx for real clients.

14.13 When to reach for third-party packages

How to decide

1. **Is the stdlib version good enough?** For scripts and tools that must run anywhere without installs (CI helpers, system scripts), prefer the standard library.
2. **Is there a clear community standard?** If a package is the de-facto choice, well maintained, and widely used, the dependency is usually worth it.
3. **Cost of a dependency:** install size, security updates, licence, and supply-chain risk (typo-squatting packages exist). Pin versions and use a lock file (Chapter 8).

Need	stdlib	Common third-party choice	Why switch
HTTP client	`urllib.request`	**requests**, **httpx**	sessions/pooling, JSON, auth, retries, async (httpx)
CLI	`argparse`	**click**, **typer**	composable commands, less boilerplate, type-hint driven (typer)
Terminal output	`print`, `logging`	**rich**	colours, tables, progress bars, pretty tracebacks and logging handler
Dates	`datetime`, `zoneinfo`	`python-dateutil`, `pendulum`, `arrow`	month arithmetic, fuzzy parsing
Validation	dataclasses	**pydantic**	runtime validation (Chapter 13)
Data tables	`csv`, `statistics`	**pandas**, **polars**	vectorised analysis at scale
Numerics	`math`, `random`	**NumPy**	fast arrays
Testing	`unittest`	**pytest**	simpler tests, fixtures, plugins (Chapter 15)
Env config	`os.environ`	`python-dotenv`, `pydantic-settings`	`.env` files, typed settings
Passwords	`hashlib.scrypt`	`argon2-cffi`, `bcrypt`	modern password hashing
Database	`sqlite3`	**SQLAlchemy**, `psycopg`	ORM / PostgreSQL

Quick taste of what the three named in the outline look like (need `pip install requests rich click` – not run here):

```python
import requests
r = requests.get("https://api.github.com/repos/python/cpython", timeout=10)
r.raise_for_status()                # HTTPError on 4xx/5xx
print(r.json()["stargazers_count"])

from rich.console import Console
from rich.table import Table
t = Table("service", "status"); t.add_row("api", "[green]up[/]")
Console().print(t)                  # a coloured, boxed table

import click
@click.command()
@click.argument("name")
@click.option("--count", default=1, type=int)
def hello(name: str, count: int) -> None:
    for _ in range(count):
        click.echo(f"Hello {name}")
```

TL;DR

Default to the standard library; switch when a well-maintained community standard removes real pain (requests/httpx, click/typer, rich, pytest, pydantic, pandas). Pin what you add.

14.14 Chapter self-check

1. Naive vs aware datetime? → Aware has `tzinfo`; store aware UTC, convert with `astimezone(ZoneInfo(...))` for display.
2. Timer for measuring code? → `time.perf_counter()`.
3. Regex function for validating a whole string? → `re.fullmatch`.
4. Why `log.info("x=%s", x)` instead of an f-string? → Formatting is skipped when disabled; aggregators group by template.
5. What should a library do about logging configuration? → Nothing beyond `getLogger(__name__)` (and optionally `NullHandler`).
6. argparse error exit code? → 2.
7. Safest way to run a command with user input? → `subprocess.run([...list...], check=True, timeout=...)`, no `shell=True`.
8. `random` or `secrets` for a password-reset token? → `secrets`.
9. `round(2.5)`? → 2 (banker's rounding).
10. Shallow copy of `{"a": [1]}`, append to original's list — does the copy change? → Yes.
11. How to store passwords? → Salted slow KDF (scrypt/pbkdf2/argon2), never plain SHA-256.
12. How to pass values into SQL safely? → `?` placeholders with a parameters tuple.
13. Why prefer requests/httpx over urllib.request? → Pooling, JSON/auth/retries, async (httpx), nicer API.

Chapter TL;DR

- **Time:** aware datetimes in UTC, `ZoneInfo` for display, ISO 8601 on the wire; `perf_counter` for durations, `monotonic` for timeouts.
- **re:** raw strings, named groups, `VERBOSE`, `finditer`, `fullmatch`, `sub` with a function; beware catastrophic backtracking; don't regex JSON/HTML/CSV.
- **logging:** `getLogger(__name__)`, configure once with `dictConfig`, lazy args, `log.exception`, libraries don't configure, structured logs in production.
- **argparse:** positionals/options/subcommands, `main(argv=None)`, exit code 2 on errors.
- **OS:** `os.environ.get`, `sys.argv` / `sys.exit`, `subprocess.run([...], check=True, timeout=...)`, `shutil` for trees, `pathlib` for paths, `tempfile` for scratch space.
- **Numbers:** `random` (seeded `Random`) vs `secrets`; `math.isclose`, banker's `round`; `statistics` mean/median/stdev/quantiles/regression.
- **Toolbox:** `deepcopy`, `pprint.pp`, `StrEnum`, `uuid4` / `uuid7`, `hashlib` (SHA-256; KDFs for passwords), `json` `default=`, `csv.Dict*`.
- **sqlite3:** placeholders, `with con:` transactions, `sqlite3.Row`.
- **HTTP:** `urllib.parse` always; `urllib.request` for quick scripts; requests/httpx for real work.
- **Third-party** when a community standard clearly beats the stdlib — then pin it.